Flight-envelope haptic controller

ENGE X200 · Assignment 2 · STM32C031C6

Student name: [to be supplied]
Student ID: [to be supplied]

1,513 prose words · AI-assisted development draft

Evidence status: analytical predictions, software validation and a reported sensor-read failure. Successful Wokwi operation and recording remain unverified.

1. Purpose, interpretation and evidence

This project implements the Assignment 2 flight-envelope haptic controller on the STM32C031C6. Pitch and roll servos indicate individual limits, while a third servo indicates combined tilt. A separate mode machine controls manual/autopilot indication. Control calculations are separated from hardware access so their state transitions can be tested independently. The assignment PDF takes precedence over the earlier System Overview: its equations and five-degree release deadband differ materially from that overview. The source is organized around small responsibilities: angle conversion, reusable region selection, servo conversion, debouncing and mode arbitration.

Nominal boundaries select stronger feedback at 30°/50° for individual axes and 20°/70° for tilt. Exact equality is assigned to the stronger region because parts of the brief omit equality. This is an AI-assisted development draft; the assignment prohibits AI-generated code and permits editing assistance only. Its eligibility therefore remains subject to course rules. No successful Wokwi demonstration or results video is claimed. A supplied version-two diagnostic log reached sensor identity and initialization, then reported an acceleration-read failure.

2. Orientation calculation and program structure

The angles_from_accel subroutine converts signed accelerometer samples to g using 16384 counts/g for the MPU6050 ±2 g range. It computes roll $\varphi = \operatorname{atan2}(a_y, a_z)$, pitch $\theta = \operatorname{atan2}(-a_x, \sqrt{a_y^2 + a_z^2})$, then tilt $\alpha = \sqrt{\theta^2 + \varphi^2}$. Angles are converted from radians to degrees before threshold comparisons. The four-quadrant roll calculation preserves signs beyond 90°; replacing atan2 with a ratio would lose quadrant information.

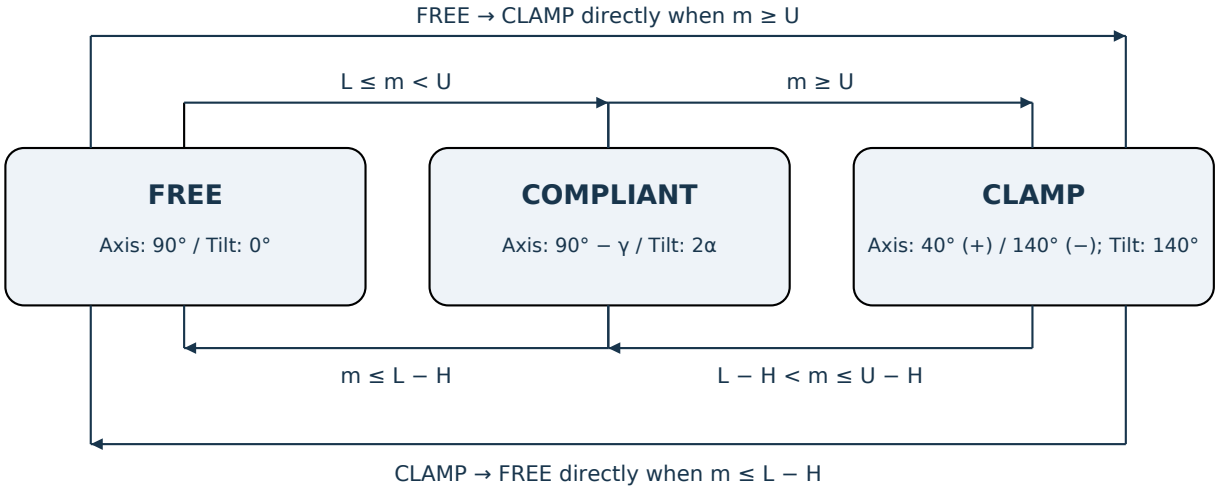
An all-zero vector has no defined gravity direction and is rejected. Non-finite or out-of-domain angles also select the neutral fault response. These equations assume gravity-dominated constant-velocity flight; sustained linear acceleration can corrupt the estimated orientation. No gyro fusion is implemented. Common acceleration scaling does not change the angles, but explicit g conversion makes test inputs interpretable. The control module uses no dynamic memory and stores independent pitch, roll and tilt region histories. Raw values are converted to floating point before squaring, avoiding overflow from multiplying signed sensor integers. The pitch denominator combines transverse acceleration components and is non-negative. A common sensor magnitude change leaves the ideal angle unchanged, although it does not remove dynamic-acceleration errors.

Every 20 ms, one sample supplies all three angles, region transitions, mode decisions and servo targets. Updating these together avoids mixing old and new readings. Servo command updates therefore match the 50 Hz PWM rate. Button sampling occurs every millisecond; its debouncer produces one event per stable press. A held input cannot toggle the mode on every control

iteration. Fault recovery temporarily interrupts normal sampling while neutral commands remain active.

3. Reusable feedback state machine

Figure 1 defines the reusable region machine. Magnitude m is |pitch| or |roll| with boundaries L = 30° and U = 50°; for tilt, m = α, L = 20° and U = 70°. Release hysteresis H is 5°. The same region_update function operates on three independent states. Stronger feedback starts at its nominal boundary; disengagement requires moving five degrees farther toward the center.



All unspecified conditions retain state. Clamp output sign follows current angle.

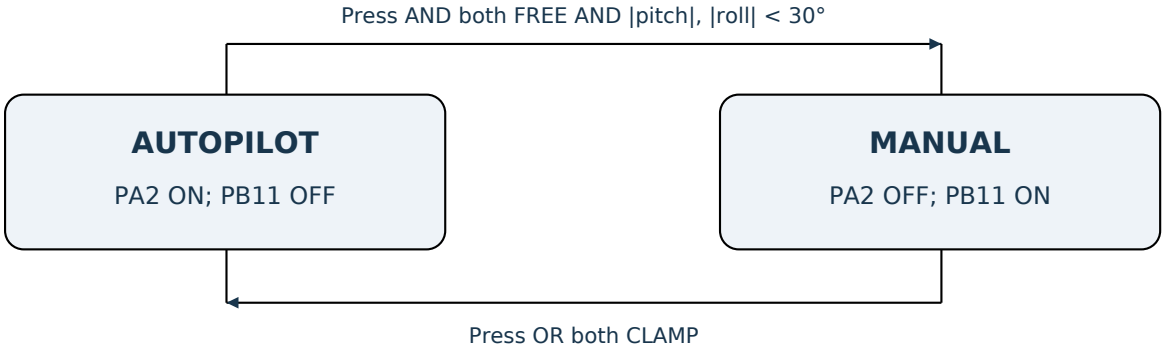
Figure 1. Reused region FSM. Axis parameters: L = 30°, U = 50°; tilt parameters: L = 20°, U = 70°. H = 5° in both cases.

FREE enters CLAMP directly when m ≥ U, otherwise COMPLIANT when m ≥ L. COMPLIANT enters CLAMP at m ≥ U and returns to FREE at m ≤ L−H. CLAMP returns directly to FREE at m ≤ L−H, otherwise enters COMPLIANT at m ≤ U−H. Other inputs retain the state. Direct jumps avoid artificial intermediate frames after large slider changes. Thus pitch releases clamp at 45° and compliance at 25°, while tilt releases at 65° and 15°. Transition ordering is significant: testing the most distant destination first permits one-sample moves from FREE directly to CLAMP and back to FREE. The clamp sign is taken from the current signed angle, allowing an abrupt positive-to-negative command to select the opposite bound without corrupting the stored region.

History matters: 28° is FREE when approached from center but COMPLIANT when approached from 40°; 47° remains CLAMP after 60°. This memory prevents chattering. FREE commands pitch/roll neutral 90° and tilt neutral 0°. Wokwi’s official servo SVG rotates clockwise as its absolute angle increases [6], while the assignment defines clockwise as negative. Consequently, compliant pitch/roll commands are 90° minus signed input. Positive clamp commands 40° and negative clamp 140°. Tilt commands 2α in compliance and 140° in clamp, with saturation. These are expected command angles, not measured servo positions.

4. Autopilot, button events and faults

Figure 2 starts in AUTOPILOT. MANUAL returns to AUTOPILOT after a button event or simultaneous pitch and roll CLAMP states. AUTOPILOT permits MANUAL only after a button event with both region states FREE and both measured magnitudes strictly below 30°. Tilt alone does not change mode. PA2 illuminates for autopilot and PB11 for manual; these outputs are mutually exclusive. Both modes retain the specified feedback law because no separate autonomous flight-control algorithm is supplied.



Startup or sensor fault → AUTOPILOT. Unsafe presses are not queued.

Figure 2. Mode FSM and LED state actions. A “press” means one debounced rising-edge event.

Checking state and angle preserves the hysteresis requirement during handover. An axis below 30° may still retain compliance; a raw value exactly at 30° is outside the strict free interval. Unsafe presses are consumed rather than queued until the stick centers. The debouncer requires an unchanged candidate level for 30 ms and emits only a rising-edge event. A release must settle before another press. Unsigned elapsed-time subtraction handles millisecond counter wraparound. Mode arbitration uses mutually exclusive branches so one button event cannot perform two transitions in the same update. Fault handling takes priority over handover. Once recovery succeeds, previously unsafe or discarded button events cannot unexpectedly engage manual mode, so the operator must make a new deliberate request.

Sensor faults force neutral outputs and AUTOPILOT, discarding stale angles and resetting region history. Initialization is retried once per second. Valid data restores feedback, but manual operation still requires a fresh safe press. The revised recovery path also covers failed configuration writes and acceleration bursts. It tries hardware transfer alternatives, then selects open-drain GPIO I2C automatically when necessary. After configuration it waits 50 ms, covering at least two configured sensor frames. Fault recovery remains an educational fallback, not an aircraft safety certification.

5. Peripheral configuration and timing

CMSIS definitions support register access, with startup initializing memory and the interrupt vector base. The linker allocates 32 KiB flash and 12 KiB SRAM. HSI supplies 48 MHz with undivided bus clocks. TIM3 prescaler 47 gives 1 µs ticks; auto-reload 19999 yields a 20 ms period. PWM mode 1 and compare preload drive channel 1/PB4 for roll, channel 2/PB5 for pitch, and channel 3/PB0 for tilt. These pins use AF1, not the earlier overview’s AF0.

The pulse conversion assumes 1000–2000 µs corresponds to 0–180°, making axis neutral 1500 µs and tilt neutral 1000 µs. That Wokwi pulse-to-angle calibration is unverified; displayed positions must be checked before treating predicted angles as observed results. Neutral compare values are prepared before enabling PWM. Real actuators would require calibrated

linkage and suitable power; positional servos only visualize the requested behavior, without demonstrating actual force or torque control. TIM3 continues generating pulses autonomously between control updates, including during sensor retries. A millisecond SysTick supports timeouts and button settling independently of the twenty-millisecond sensor schedule. Diagnostic output uses an otherwise unused serial pin and does not share the servo or sensor connections specified in the assignment.

I2C1 uses open-drain PB8/PB9 AF6 with external pull-ups. TIMINGR 0x00B4274F targets below 400 kHz at 48 MHz; physical edge timing requires measurement. The MPU6050 does not support 1 MHz Fast-mode Plus. Initialization verifies WHO_AM_I = 0x68, wakes the device and selects ±2 g. Reads obtain acceleration registers 0x3B–0x40 together. The GPIO fallback checks real acknowledgements and reads six bytes before final NACK/STOP, with bounded clock-stretch waits; it never substitutes acceleration values. PA12 uses pull-down, and PA9 transmits diagnostics.

6. Test method, results and remaining evidence

Table 1 constructs unit-gravity vectors from selected θ and φ using ax = −sin θ, ay = cos θ sin φ, az = cos θ cos φ. Their magnitude is one g before rounding. Integer sampling and slider fidelity allow small angle differences. Independent cases begin with centered input unless prior history is stated. Pitch-only and roll-only refer to isolated orientation axes; their 40° inputs also activate tilt feedback, correctly commanding 80° there. The construction also makes combined-clamp testing physically consistent with a one-g input: negative X supplies positive pitch, while positive Y and Z establish positive roll. Resetting before independent table cases prevents retained hysteresis from changing their predicted initial states. Release sweeps intentionally preserve history instead.

Case / initial state	ax, ay, az (g)	θ / φ / α (°)	Pitch / roll / tilt (°)
Centered / reset	-0.00000, 0.00000, 1.00000	0 / 0 / 0.00	90 / 90 / 0.00
Pitch compliant / reset	-0.64279, 0.00000, 0.76604	40 / 0 / 40.00	50 / 90 / 80.00
Roll compliant / reset	-0.00000, 0.64279, 0.76604	0 / 40 / 40.00	90 / 50 / 80.00
Tilt only / reset	-0.30902, 0.29389, 0.90451	18 / 18 / 25.46	90 / 90 / 50.91
Pitch negative / reset	0.86603, 0.00000, 0.50000	-60 / 0 / 60.00	140 / 90 / 120.00
Roll negative / reset	-0.00000, -0.86603, 0.50000	0 / -60 / 60.00	90 / 140 / 120.00
Both clamp / MANUAL	-0.86603, 0.43301, 0.25000	60 / 60 / 84.85	40 / 40 / 140.00

Table 1. Analytically predicted vectors and servo commands, not observed Wokwi measurements. All vectors have unit magnitude before rounding. “Reset” means FREE region history.

With both axes at 18°, each individual axis stays FREE while tilt is 25.46°, commanding approximately 50.91°. This is the required tilt-only feedback case. Simultaneous 60° pitch and roll give tilt 84.85°, both individual commands 40° and tilt 140°, forcing MANUAL to AUTOPILOT. Negative 60° verifies the opposite 140° bound. Sweep each axis through 60°, 47°, 45°, 28°, 25° to demonstrate CLAMP, CLAMP, COMPLIANT, COMPLIANT, FREE; tilt uses 75°, 67°, 65°, 18°, 15°.

The clean GNU Arm GCC 14.2.1 build passed strict C11 warnings, producing ELF/BIN/HEX artifacts. Firmware uses 12,760 flash bytes and 156 static RAM bytes; stack usage is unmeasured. Host validation passed 7,376 control checks, 65 GPIO protocol checks and 198 recovery checks exercising actual initialization/read routines through mocked transports. Control and protocol suites also passed address/undefined-behavior sanitizers. These checks establish software consistency, not successful emulation. The final recording must show sensor inputs,

servo behavior, LEDs and safe/rejected button transitions. No successful Wokwi run or results video exists yet.

References and attribution

[1] Adelaide University. ENGE X200 Assignment 2, supplied PDF.

[2] STMicroelectronics. STM32C031 datasheet; RM0490. github.com/STMicroelectronics/cmsis-device-c0.

[3] Arm. CMSIS_5. github.com/ARM-software/CMSIS_5.

[4] InvenSense. MPU6050 Product Specification rev. 3.4; Register Map rev. 4.2.

[5] Wokwi documentation: docs.wokwi.com. ST pin maps: github.com/STMicroelectronics/STM32_open_pin_data.

[6] Wokwi. servo-element.ts:85, commit 5178c0cc1b455719fdf562169293797e29d73dac. github.com/wokwi/wokwi-elements.

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